

Solver-Aware Neural Dynamics

Interrogating Integrators and Basis Functions in Kolmogorov–Arnold Network ODEs

COURSE

CSE 402: Numerical Analysis, Simulation & Modeling

PRESENTATION

Project Proposal Presentation

DATE

17 August 2026

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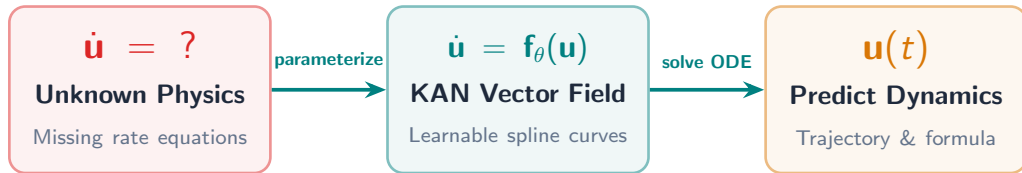
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Base Paper: Learning the Missing Physics



Base Paper: Z. Koenig, J. Kim, Y. Deng, *KAN-ODEs: Kolmogorov–Arnold Network Ordinary Differential Equations for Learning Dynamical Systems and Hidden Physics*, CNAME / arXiv:2407.04192 (2024), MIT.

Learning Dynamic System
Hidden Equation Explore

Base Paper's Method: Three Key Decisions

1. Integrator

Forward Integration Pass

- **Tsit5** adaptive RK 5/4
- Continuous-time forward state
- Grid-independent evaluation

2. Activation

KAN Architecture

- Learnable 1D edge curves
- **Gaussian RBF** basis
- Symbolic law discovery

3. Optimization

Backward Pass & Training

- **Adjoint Sensitivity** gradients
- **Adam** with L_1 sparsity
- Memory-efficient backprop

10× Fewer Epochs

Lotka–Volterra: 10^4 vs 10^5 epochs

Parameters: **240** (KAN) vs 252 (MLP)

Our Extension: Stress-Testing the Design Choices

The base paper fixed **Tsit5** + **Gaussian RBF**. We systematically interrogate both choices.

1. Solver Ablation

maps to Integrator

- Forward Euler ($p = 1$)
- Heun RK2 / Classical RK4
- Adaptive Dormand–Prince

Metrics: Loss · NFE · Drift

2. Basis Swap

maps to Activation

- Cubic B-splines ($k = 3$)
- Newton Polynomial
- Lagrange Polynomial

Metrics: Accuracy · Params

3. Robustness

new beyond the paper

- Damped Pendulum dynamics
- SIR Epidemic dynamics
- Noise sweep ($\sigma \leq 0.10$)

Metric: Generalization

Controlled Setup: Adjoint gradients and Adam optimizer remain fixed, strictly isolating solver and basis effects.

Expected Methodology: One Reproducible Pipeline



Results Deliverable:

- Train/Extrapolation MSE · NFE per step
- Sensitivity to noise ($\sigma \leq 0.1$) · Wall-clock speed

Numerical Techniques Explore:

Euler/RK ODE integration · Spline & polynomial interpolation · Truncation error · Optimization